Hazard Analysis
Squigglr

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Table 1: Revision History

Date	**Developer(s)**	**Change**
14 October 2026	Rayan Nasrallah	Added critical assumptions CA-001 to CA-006 and hazards H-RAY-001 to H-RAY-007 (#91).
15 October 2026	Rayan Nasrallah	Added my reflection responses to the appendix and my generative AI use disclosure (#140).

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[You are free to modify this template. —SS]

1 Introduction

[You can include your definition of what a hazard is here. —SS]

2 Scope and Purpose of Hazard Analysis

[You should say what **loss** could be incurred because of the hazards. —SS]

3 System Boundaries and Components

[Dividing the system into components will help you brainstorm the hazards. You shouldn't do a full design of the components, just get a feel for the major ones. For projects that involve hardware, the components will typically include each individual piece of hardware. If your software will have a database, or an important library, these are also potential components. —SS]

4 Critical Assumptions

The assumptions below are kept few and are written so that they do not remove hazards. Squigglr does not assume that an adult watches the child during practice, that the network is always available, or that the device meets Squigglr's requirements; those conditions are treated as failure modes in the Failure Mode and Effect Analysis (FMEA). Sources are cited by their identifiers in the Software Requirements Specification (SRS), Appendix A, and interview turns by speaker and timestamp, for example (SRC-005, P1, 34:13). Each assumption lists the H-RAY hazards it affects.

CA-001 An adult sets up Squigglr for a child, chooses and starts the lesson and its starting stage, and is the only person meant to accept a suggestion to go back a stage. Within a lesson the child does not navigate, exit or restart; Squigglr moves between stages. An adult may not be present or watching while the child practises.
Rationale: children should practise without constant adult help, while adults handle setup (SRC-005, P1, 34:13), and adults decide progression (SRC-005, P1, 28:37; CON-007).
Hazards affected: H-RAY-001, H-RAY-004, H-RAY-005.

CA-002 The device, browser, touchscreen or stylus in use may not be one of the supported combinations (`SUPPORTED_DEVICES`, TBD), and Squigglr may be opened on a device without a stylus, where the child can draw with a finger but not as intended (ASM-003).

Rationale: Squigglr is a website opened from a web address with no installation (CON-001, CON-002), so the team does not control the device. The supported combinations are not yet defined (CON-003), and the team did not promise that Squigglr works on every device (SRC-005, Moustafa, 6:03).
Hazards affected: H-RAY-002.

CA-003 The network connection may be slow or lost at any time, including during an attempt.
Rationale: attempts are saved to an account and reviewed later from another device, and offline access was not promised (SRC-005, Moustafa, 6:03). Home and school networks are outside the team's control.
Hazards affected: H-RAY-002, H-RAY-006.

CA-004 One device may be used by several children, at school or at home.
Rationale: school devices may be shared by several children (SRC-004), and Squigglr may be used at home, at school or in other settings such as rehabilitation centres (SRC-005, P2, 21:25).
Hazards affected: H-RAY-006.

CA-005 The adults who read practice records are not assumed to have training in visual-motor development, and they may show the records to other people, such as teachers or therapists.
Rationale: parents and caregivers are not assumed to be specialists (SRS, Hands-On Users), and records exist to support conversations about what a child needs (SRC-005, Moustafa, 6:03). Squigglr makes no diagnosis or treatment claims (SRC-001).
Hazards affected: H-RAY-003.

CA-006 The supporting adult chooses the adult-set values, such as `STAGE_PASS`, `DIFFICULTY_STREAK` and `ATTEMPT_LIMIT`, for each child within a range the team sets (SRC-008). The ranges are not found from child testing, and the value an adult picks, or the default, may not suit their child.
Rationale: no information is gathered from children and no study with children is run (CON-005, CON-006), so the ranges cannot be tuned on children (SRS, Goals of the Project). Children reach these skills at different points (SRC-005, P1, 20:27).
Hazards affected: H-RAY-004, H-RAY-005, H-RAY-007.

5 Failure Mode and Effect Analysis

[Include your FMEA table here. This is the most important part of this document. —SS] [The safety requirements in the table do not have to have the prefix SR. The most important thing is to show traceability to your SRS. You might trace to requirements you have already written, or you might need to add new requirements. —SS] [If no safety requirement can be devised, other mitigation

strategies can be entered in the table, including strategies involving providing additional documentation, and/or test cases. —SS]

5.1 Supervision, Context of Use and Use of Records (H-RAY)

Table 2 covers hazards from how Squigglr is supervised, where and on what it is used, and how its records are read. Each recommended action starts with Eliminate, Mitigate or Document/Test. Mitigating requirements trace to the goals (G.#) and constraints (CON-###) in the SRS, or to safety requirements SAF-RAY-### in the SRS, Safety-Critical Requirements.

Table 2: FMEA: supervision, context of use and use of records

Hazard ID	**Component**	**Failure mode**	**Cause(s)**	**Effect and loss**	**Category**	**Recommended action**	**Mitigating requirement IDs**
H-RAY-001	Lesson flow; adult controls	A child practises with no adult watching, or the adult is inattentive, and the child reaches an adult-only action.	Children are meant to practise without constant adult help (CA-001); adult controls use the same touch interface as the child.	The child accepts a suggestion to go back a stage, or changes lesson or stage, without an adult's decision; the child's distress goes unnoticed. Loss: progression not chosen by an adult; well-being.	Psychological/ well-being	Mitigate: place every adult-only action (choosing a lesson or stage, accepting a suggestion) behind an adult check that a child cannot pass by tapping icons; keep a pending suggestion waiting, without blocking practice, until an adult reviews it.	CON-007, G.3, SAF-RAY-001
H-RAY-002	Browser front end; saving of attempts	Squigglr is used on an unsupported device or browser, without a stylus, or loses its connection during an attempt, at home, at school or on a shared device.	Device not controlled by the team (CA-002); no offline access (CA-003).	Strokes are not captured correctly, the child's attempt is lost, or a lesson cannot be completed; the child is frustrated and the record is incomplete. Loss: lost practice data; availability.	Data integrity or Availability	Mitigate: before a lesson starts, check the device and browser against `SUPPORTED_DEVICES` and show the adult a notice when they do not match; keep each attempt until the server confirms it is saved, and tell the adult when an attempt could not be saved.	CON-001, CON-002, CON-003, G.5, SAF-RAY-002, SAF-RAY-003

Hazard ID	Component	Failure mode	Cause(s)	Effect and loss	Category	Recommended action	Mitigating requirement IDs
H-RAY-003	Attempt records; progress summaries	Practice records are used as clinical or diagnostic evidence, or as a measure of the child's ability.	Adults reading records lack specialist training, and records may be passed to others (CA-005); fewer attempts may be read as better skill (SRC-005, Moustafa, 6:03).	An adult or professional draws a wrong conclusion about the child, such as a label or a placement decision. Loss: harm to the child's well-being and to decisions about support.	Psychological/ well-being	Eliminate: show no diagnostic labels, judgements of the child's abilities or comparisons with other children in any record or summary. Mitigate: state on every summary and export that the records describe practice only and are not an assessment.	G.5, G.6, SAF-RAY-004
H-RAY-004	Lesson flow; feedback	Repeated failed attempts at one stage continue until the child becomes frustrated or agitated.	Failure feedback repeats; `ATTEMPT_LIMIT` may be too high for a given child (CA-006); an improvement extension lengthens the lesson; no adult may be watching (CA-001).	The child becomes distressed, stops practising or refuses the activity later. Loss: well-being; the practice benefit.	Psychological/ well-being	Mitigate: end the lesson at `ATTEMPT_LIMIT`, or at `ATTEMPT_LIMIT_MAX` when extended, and save progress (G.3); let audio feedback be muted (SRC-005, P2, 37:39). Document/Test: test that each lesson ends at `ATTEMPT_LIMIT` and that no extended lesson passes `ATTEMPT_LIMIT_MAX`.	G.3, SAF-RAY-005

Hazard ID	Component	Failure mode	Cause(s)	Effect and loss	Category	Recommended action	Mitigating requirement IDs
H-RAY-005	Lesson flow; attempt records	After a lesson ends at `ATTEMPT_LIMIT`, it is restarted at once and the repetition continues.	No adult may be watching (CA-001); a restart starts a new count of attempts.	The limit does not stop the repetition, and the adult cannot see how often it happened. Loss: well-being; misleading records.	Psychological/ well-being	Mitigate: record each lesson that ended at `ATTEMPT_LIMIT` and show the adult how many times this happened per lesson; keep earlier attempts when a lesson is restarted. Document/Test: test that a restart after the limit keeps the earlier record.	G.3, G.5, SAF-RAY-006
H-RAY-006	Learner selection; attempt records	On a shared device, attempts or a pending suggestion are saved to, or shown for, the wrong child.	Several children use one device (CA-004); the active child is not changed between sessions; a lost connection leaves a session open (CA-003).	One child's records are shown to another child or adult, or a child's history is mixed with another's, so an adult accepts a suggestion based on the wrong attempts. Loss: privacy; record integrity.	Privacy/ security	Mitigate: show which learner is active on every practice screen in a way a child who cannot read can recognize; require an adult action to switch learners; show the attempts behind a suggestion when the adult reviews it.	G.5, SAF-RAY-001, SAF-RAY-007

Hazard ID	Component	Failure mode	Cause(s)	Effect and loss	Category	Recommended action	Mitigating requirement IDs
H-RAY-007	Child profile settings; lesson flow	A stage check or lesson limit uses a value that does not suit the child: `STAGE_PASS` too strict or too lenient, or `ATTEMPT_LIMIT` not larger than `DIFFICULTY_STREAK`, or `ATTEMPT_LIMIT_MAX` below `ATTEMPT_LIMIT`.	The adult misjudges the example drawings or leaves a default that does not suit the child (CA-006); a value is saved outside its range; a child changes a setting.	A child cannot pass a stage they can draw, or passes one they cannot, and the records mislead the adult. Loss: well-being; misleading records.	Psychological/ well-being	Mitigate: accept only values within the team-set range, and only an `ATTEMPT_LIMIT` larger than `DIFFICULTY_STREAK`; show example drawings of each end of the `STAGE_PASS` range; place settings behind the adult check; show the values in use next to the child's records. Document/Test: test that out-of-range and invalid values are refused and that the default is the midpoint.	G.2, G.3, SAF-RAY-008

6 Safety and Security Requirements

[Newly discovered requirements. These should also be added to the SRS. (A rationale design process how and why to fake it.) —SS]

7 Roadmap

[Which safety requirements will be implemented as part of the capstone timeline? Which requirements will be implemented in the future? —SS]

8 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

8.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

8.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

8.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

Individual Disclosures

Rayan: I used Anthropic's Claude Opus 5.5 at medium effort, accessed through Claude Code, while preparing my sections of this document. I developed the ideas and made the project decisions in my sections myself, and Moustafa and I conducted the specialist interview and follow-up questionnaire. A major use of AI was organizing our sources: I used the model to sort through the interview transcript, questionnaire answers, and meeting notes, and to extract the specific statements, with their timestamps, that supported the ideas I wanted to include. This would have been very difficult to do by hand given the length of

the material. I also used it to turn my notes and decisions into draft wording, to improve sentence structure, clarity, and flow, and to help with LaTeX formatting. Through Claude Code, it also helped manage the repository (branches, commits, and merges) and helped me review my teammates' pull requests by checking their changes against our decisions and the rest of the document; I decided which review comments to post.

All ideas and decisions in my sections are my own. I reviewed every AI-assisted passage against my intended meaning and checked each cited statement against the original transcript, questionnaire, or meeting notes to confirm that it says what we claim it says.

9 Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

   Rayan: Identifying the hazards was easier than expected because we already had good sources to work from. The transcript of our interview with three domain specialists on October 1, 2026 gave us concrete statements, with timestamps, about how the intended children use apps and what could go wrong for them. For example, the specialists said adults should decide progression rather than the app, that audio feedback should be possible to mute, and that Squigglr may be used at home, at school or in other settings. Each of these pointed directly to a failure mode we could analyse, so we were not guessing at hazards from scratch. Our meeting with our supervisor, Dr. Anand, on October 14, 2026 helped in the same way: once Dr. Anand explained that thresholds should be set by the adult for each child within a range, the hazards around wrong or extreme threshold values became clear. Because the SRS already recorded these sources, business rules and assumptions, we could trace each hazard and critical assumption back to a specific statement, which also made the mitigations easier to justify.

2. What pain points did you experience during this deliverable, and how did you resolve them?

   Rayan: The main pain point was that my hazards kept changing as other parts of the documents changed. I wrote my first set of hazards and critical assumptions based on the SRS as it stood at the time, where the team would set fixed values for thresholds such as `DIFFICULTY_STREAK` and `ATTEMPT_LIMIT`. In our meeting on October 14, 2026, Dr. Anand advised that suitable values differ from child to child, so I changed those thresholds to be set by the supporting adult within a range the team defines. When I reviewed the hazard analysis again, I found that this changed the hazards: the risk was no longer only a wrong team-chosen value, but also an adult choosing a value at the edge of the range, so I

added a new hazard and safety requirement for it. The same thing happened when I added the improvement rule, which can extend a lesson while a child is improving, and when we confirmed that the child does not navigate inside a lesson. Each time, I had to go back and update the affected hazards, critical assumptions and mitigations so they matched the SRS. It was time-consuming, but it showed me how closely the hazard analysis depends on the requirements, and that it has to be reviewed again whenever the SRS changes.

3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?

   Rayan: Before this deliverable, the team had already thought of two of my risks. Because Squigglr runs in a browser, we expected that attempts could be lost or not saved on an unreliable browser or device (H-RAY-002). We also knew that adults might read a child's records as a diagnosis (H-RAY-003), which is why the problem statement already avoids any diagnostic claims.

   The other five risks I only thought of while doing the hazard analysis. They came about because this was the first time I had to work out the requirements for the application in detail, while writing the SRS alongside this document. Most of them came from the critical assumptions, and several of those assumptions came from our interview with the specialists on October 1, 2026. The specialists said children can use the app through images and icons while adults handle setup, which led to the assumption that an adult sets up each session. Asking what happens if no adult is actually watching led to the risk of a child reaching adult-only actions (H-RAY-001), and, together with defining what happens after repeated failed attempts, to the risks of a child becoming distressed (H-RAY-004) and of a lesson being restarted straight after it ends (H-RAY-005). The specialists also said Squigglr may be used at home, at school or in other settings, which led to the assumption that a device may be shared and to the risk of a child's records being saved under the wrong profile (H-RAY-006). The last risk came from our meeting on October 14, 2026, where Dr. Anand advised that adults should set the thresholds for each child; this raised the risk of a threshold being set to a value that does not suit the child (H-RAY-007).

   Looking back, most of my risks only became visible once the requirements were concrete enough to question. The risks we knew about beforehand were general ones that apply to almost any browser app, while the ones I found during this deliverable were specific to how children and adults will actually use Squigglr. This showed me that the hazard analysis cannot be done well in isolation; it depends on knowing the requirements and on hearing from people who understand the users.

4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?

   Rayan: Most software carries risks that have nothing to do with physical harm. One is privacy and security. Almost every application stores some information about its users, and if that information is exposed, accessed by the wrong person or mixed up between users, it can harm those users and destroy their trust in the product. This matters even more when the users are vulnerable. In Squigglr, for example, the records describe a child's difficulties, so they must only be visible to the adults responsible for that child. A second is data integrity and availability. People make decisions based on what software shows them, so data that is lost, corrupted or shown incorrectly can lead them to the wrong decision, even when nothing visibly crashes. In our case, an adult choosing a child's next lesson relies on the attempt records being complete and accurate. A third is psychological harm. Software that frustrates, confuses or misleads its users can affect their well-being or lead them to wrong conclusions, for example by implying something about a person that the software cannot actually judge. For Squigglr, that means avoiding repeated failure without a break and making sure progress summaries are never read as a diagnosis. These risks are important to consider because they are easy to overlook when there is no obvious physical danger, yet they can cause real harm to users and to the people who rely on the software.